Noakhali Science and Technology University





UniMarket

SPL-2 Final Project User Manual

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1. Project Overview

1.1 Purpose of the Project

UniMarket is a hyper-local, campus-based digital marketplace designed to make academic life easier and more affordable. Its primary purpose is to provide a trusted digital ecosystem where university students can securely buy, sell, rent, or exchange academic resources, electronics, and daily essentials with their peers within the same institution.

1.2 Main Objectives

- To significantly reduce the financial burden on students by promoting the reuse and sharing of academic materials.
- To create a secure, scam-free environment by ensuring all listed items undergo strict admin moderation before becoming public.
- To introduce flexible transaction methods, enabling students not just to buy, but also to rent items for short-term projects or exchange items without spending money.

1.3 Target Users

- **Primary Users:** University students (both buyers and sellers) looking for affordable textbooks, renting lab equipment, or wanting to earn money by selling items they no longer need.
- **Secondary Users:** Campus Administrators who oversee the platform, moderate product uploads, and resolve user disputes.

1.4 Major Functions of the System

From a user's perspective, UniMarket offers a seamless experience with the following major functions:

- **Tri-Mode Listing:** Users can easily snap a photo of a product and list it for Sale, Rent, or Exchange directly from their mobile phones.
- **Smart Browsing & Cart:** Users can search for specific items using categories, add products to their cart, and proceed to a streamlined checkout.
- **Real-Time Communication:** A built-in chat system allows buyers and sellers to negotiate prices and coordinate safe physical meetups on campus.

- **Instant Notifications:** Users receive real-time alerts whenever they get a new order, a rent request is accepted, or an admin approves their product.
- **Dedicated Support:** A formal ticketing system allows users to report issues and get quick help directly from the platform administrators.

2. System Requirements

2.1 Hardware Requirements

To smoothly run the UniMarket application, the following minimum or recommended hardware is required:

- **Device:** A standard Smartphone or Tablet.
- **RAM:** Minimum 5 GB of RAM (Recommended: 6 GB or higher for optimal image loading).
- **Storage:** At least 500 MB of free storage space to install the app and store temporary cache data.
- **Internet Connection:** An active high-speed internet connection (Wi-Fi or 3G/4G/5G Cellular Data) is strictly required to sync with the real-time database.

2.2 Software Requirements

The following software platforms and system permissions are required to operate the application:

- **Mobile Operating System:** Android 6.0 (Marshmallow) or higher, OR iOS 11.0 or higher.
- **System Permissions:** The user must grant Camera and Local Storage (Gallery) permissions to upload product images, along with background network access.
- **Required Packages:** No third-party web browsers or extra plugins are needed since it is a standalone mobile application.

3. Installation and Setup

To access and use the UniMarket application, users must follow a straightforward onboarding process. The system has been designed to be highly intuitive, ensuring a smooth setup for university students.

Step 1: Download and Install the Application

Currently, the application is distributed as an Android APK (Android Package Kit) file.

- Navigate to the provided repository or drive link to download the UniMarket.apk file.
- Tap on the downloaded file. If prompted by your device, navigate to settings and enable the "Install from Unknown Sources" permission.
- Click **Install** and wait for the installation process to finish.

Step 2: Configure Necessary Settings

- Upon launching the app for the first time, ensure your device is connected to a stable internet connection (Wi-Fi or Cellular Data).
- When prompted by the operating system, grant the necessary permissions, specifically allowing the app to access your device's storage/gallery (which is required later for uploading product images).

Step 3: Create an Account

- If you are a new user, tap on the **"Sign Up"** button on the initial welcome screen.
- Fill in the required registration fields: Full Name, Email Address, Phone Number, and a secure Password.
- Click "Register" to create your account. Alternatively, users can simply tap the **"Continue with Google"** button for a faster, one-tap registration process.

Step 4: Log in to the System

1. If you already possess an account, simply enter your registered email and password on the Welcome Back screen.
2. Tap the Login button. The system will securely authenticate your credentials via Firebase Authentication and grant you access.

Step 5: Start Using the Application

- After a successful login, you will be automatically redirected to the main Home Screen (Global Feed).
- From here, you can start using the application: browse available products, use the search bar, filter by categories (e.g., Textbooks, Electronics), or tap the "+" (Sell) icon on the bottom navigation bar to upload your own items for sale, rent, or exchange.

4. Getting Started

This section outlines the basic workflow for a new user to smoothly onboard and start utilizing the UniMarket application.

4.1 Opening the Application

Once the application is successfully installed on your mobile device, locate the UniMarket icon in your device's app drawer or home screen. Tap the icon to launch the application. You will be greeted by a brief splash screen featuring the UniMarket logo, which quickly transitions into the authentication page.

4.2 Registration Procedure

To participate in the marketplace and ensure a secure ecosystem, new users must create an account:

1. On the welcome screen, tap the **"Sign Up"** link located at the bottom.
2. Fill out the registration form by providing your Full Name, University Email Address, Phone Number, and creating a strong Password.

3. Tap the **Sign-Up** button.
4. *Alternative Method:* For a faster onboarding experience, you can tap the **"Continue with Google"** button to register automatically using your existing Google account credentials.

4.3 Login Procedure

If you already have an account, or have just completed the registration process:

1. On the **Welcome Back** screen, enter your registered Email Address and Password into the designated text fields.
2. Tap the **Login** button.
3. The system will securely authenticate your credentials against the database and grant you access to the main platform.

4.4 Home Page Description

Upon successful login, you will land on the **Home Screen**, which serves as the central hub of UniMarket:

- **Search & Categories:** Located at the top, a search bar and category chips (e.g., Textbooks, Electronics) allow you to quickly filter and find specific items.
- **Global Feed:** The main body of the screen displays a scrolling grid of "Recent Uploads". These are all the products that have been successfully reviewed and approved by administrators.
- **Notification Center:** A bell icon located at the top right corner alerts you to any new messages, incoming rent requests, order updates, or admin replies.

4.5 Navigation Instructions

Navigating through the UniMarket application is primarily done using the intuitive **Bottom Navigation Bar**, which provides quick access to five core sections:

1. **Home:** Returns you to the main product feed from anywhere in the app.
2. **Chat:** Opens your messaging inbox, displaying all ongoing conversations with other buyers or sellers regarding specific products.

3. **Sell (+):** The central action button that opens the "Post an Item" form, allowing you to seamlessly upload your own products for sale, rent, or exchange.
4. **Wishlist:** Displays a curated list of all the products you have saved or "hearted" for future reference.
5. **Profile:** Opens your personal dashboard where you can edit your profile details, track your active orders/rentals, access the seller dashboard to manage your inventory, and reach out to the support ticketing system.

5. User Roles

The UniMarket system operates on a role-based architecture to ensure security, content moderation, and smooth transactions. The system explicitly defines two distinct user roles, each with its own set of permissions and a dedicated application interface.

5.1 Administrator

The Administrator has the highest level of authority and uses a separate, dedicated "Admin Dashboard" application to oversee the entire platform. Their primary role is to maintain the integrity, safety, and smooth operation of the marketplace. The features available to an administrator include:

- **Product Moderation:** Admins have the sole authority to review every newly uploaded product. They can approve genuine academic items to make them visible on the global feed, or reject spam/inappropriate items that violate campus policies.
- **User Oversight & Management:** Admins can view a comprehensive list of all registered users operating on the platform. They also have the authority to suspend or permanently ban accounts of users who engage in fraudulent activities, violate platform policies, or repeatedly upload inappropriate items.
- **System Analytics:** Admins have access to high-level statistical overviews on their dashboard, including the total number of active users, live products, pending approvals, and overall platform activities.

- **Support & Dispute Resolution:** Admins can access the Support Ticketing System to read user complaints, reply to issues, and mark tickets as "Resolved," ensuring a reliable and accountable support ecosystem.

5.2 Registered User (Buyer & Seller)

A Registered User is a verified university student who accesses the platform via the primary "UniMarket User App". In this system, every registered user acts dually as both a Buyer and a Seller, eliminating the need to create separate accounts. The features available to a registered user include:

- **As a Buyer/Renter:** Users can browse the global feed, search for specific items using filters, add products to their cart for a direct purchase, send custom rent requests for a specific duration, or propose an item exchange. They can also save items to their wishlist and track their live order statuses.
- **As a Seller:** From the internal "Seller Dashboard", users can post new items by uploading images, setting conditions, and defining prices. They can manage their active inventory, evaluate and accept/reject incoming rent or exchange requests, and track their total earnings.
- **Communication & Support:** All registered users have access to a secure, real-time chat system to negotiate with peers. Furthermore, they can raise formal support tickets directly to the administration if they face any difficulties during a transaction.

6. Functionalities and Usage Instructions (User Manual)

This section serves as the primary User Manual for the UniMarket platform, providing comprehensive, step-by-step operational instructions for both student users and administrative staff. The system is architected into two distinct client applications: the User Application (for student buyers, sellers, renters, and traders) and the Admin Dashboard Application (for platform moderation, oversight, and dispute resolution). In accordance with professional software documentation standards, each interface screen is presented with precise numbered callout indicators, accompanied by detailed functional descriptions of each interactive element and operational workflow.

6.1 User Application Manual (For Buyers & Sellers)

The User Application provides an end-to-end marketplace tailored specifically for the university campus ecosystem. Students can register with their institutional credentials, authenticate securely, recover forgotten credentials, browse categorized products, list items across flexible models (Sell, Rent, Exchange), conduct secure digital transactions, coordinate rentals and barter trades, manage personal listings, chat in real-time, submit support tickets, and update profile settings.

6.1.1 User Registration (Sign Up)

New users must create an account using their official university email credentials to gain access to the campus-exclusive marketplace. The registration procedure validates institutional membership, verifies email formatting, and establishes a secure profile.

Operational Procedure:

1. Launch the UniMarket mobile application and tap "Sign Up" at the bottom of the welcome screen.
2. Enter your Full Name in the provided text field.
3. Input your institutional email address (e.g., student@student.nstu.edu.bd).
4. Set a strong password (minimum 6 characters) and re-enter it in the confirmation field.
5. Tap the "Sign Up" button to register and proceed to the authentication state.

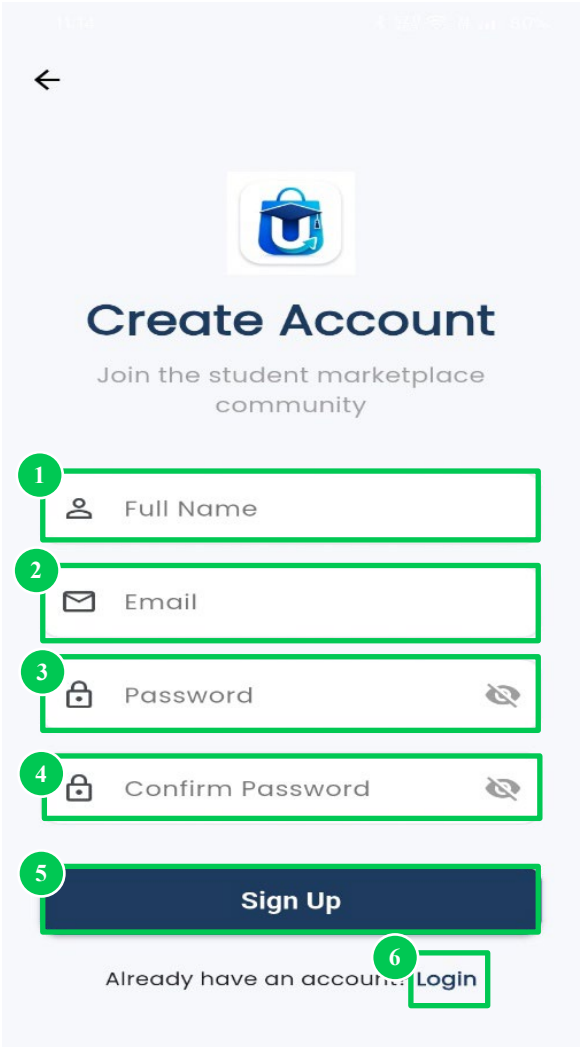


Figure 6.1: User Registration (Sign Up) Screen

1. Full Name Field: Text input box for entering the student's complete legal/academic name for identity verification.

2. Institutional Email Field: Input field restricted to verified university domain addresses (@student.nstu.edu.bd) to enforce campus community exclusivity.

3. Password Field: Secure masked field for defining a confidential account access password.

4. Confirm Password Field: Secondary password field to ensure exact character matching and eliminate typographical errors.

5. Sign Up Button: Primary action button that validates all input parameters and registers the account in Firebase Authentication.

6. Login Redirection Link: Direct navigational hyperlink enabling registered users to quickly return to the login interface.

6.1.2 User Login & Authentication

Existing registered students can log into UniMarket using their institutional email and password combination, or via seamless Google Single Sign-On (SSO).

Operational Procedure:

1. Open the UniMarket app to view the "Welcome Back" login interface.
2. Enter your registered institutional email and password in their respective fields.
3. Tap "Login" to authenticate and access the marketplace dashboard.
4. Alternatively, tap "Continue with Google" to authenticate instantly with your linked institutional Google account.

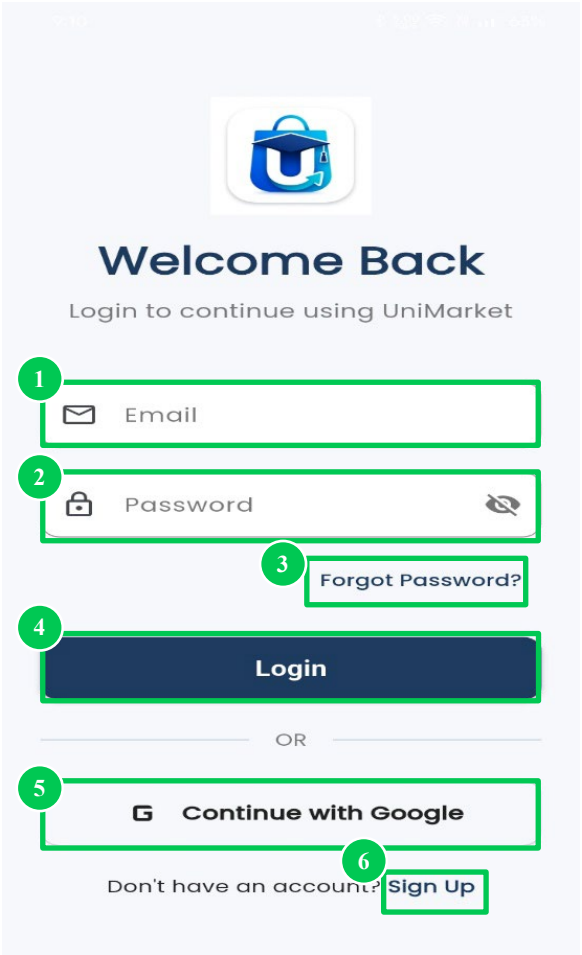


Figure 6.2: User Login Screen

1. Email Input Field: Text field for entering the student's registered institutional email address.

2. Password Input Field: Masked input box for typing the confidential account password.

3. Forgot Password Link: Action link that triggers the self-service password recovery modal when credentials are forgotten.

4. Login Button: Primary authentication button that validates credentials against Firebase Auth and launches the Home Feed.

5. Continue with Google Button: One-tap OAuth 2.0 single sign-on (SSO) button leveraging Google Identity Services for instant authentication.

6. Sign Up Redirection Link: Navigation link directing unauthenticated new visitors to the account creation form.

6.1.3 Password Recovery (Forgot Password Flow)

In case a student forgets their login password, UniMarket includes an automated, secure self-service password recovery mechanism powered by Firebase Authentication.

Operational Procedure:

1. Tap the "Forgot Password?" link on the main login screen to open the recovery dialog.
2. Input your registered institutional university email address into the prompt field.
3. Tap "Send Reset Link" to dispatch the recovery email.
4. Open your institutional email inbox, locate the message from UniMarket, and click the password reset link.
5. Enter and confirm your new password on the secure web page to restore full account access.

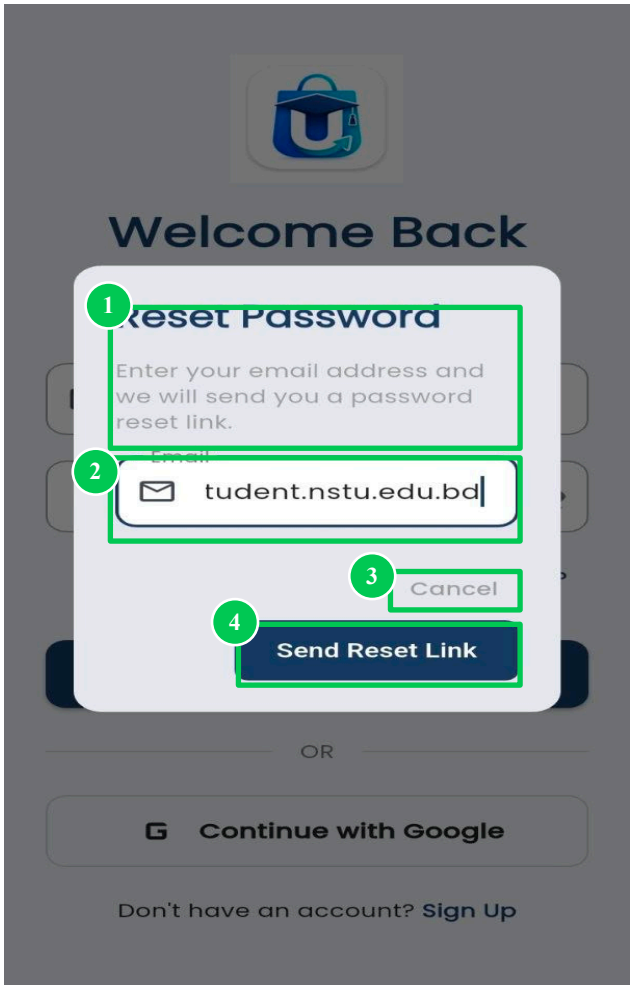


Figure 6.3: Password Reset Request Modal

1. Instruction Banner: Header banner explaining the recovery mechanism and informing the user about the email delivery.

2. Email Address Input Field: Dedicated input field for entering the registered email address to receive the password reset token.

3. Cancel Button: Dismissal control that closes the recovery dialog and returns the user to the login screen.

4. Send Reset Link Button: Execution button that triggers Firebase Authentication to generate a secure reset token and send the recovery email.

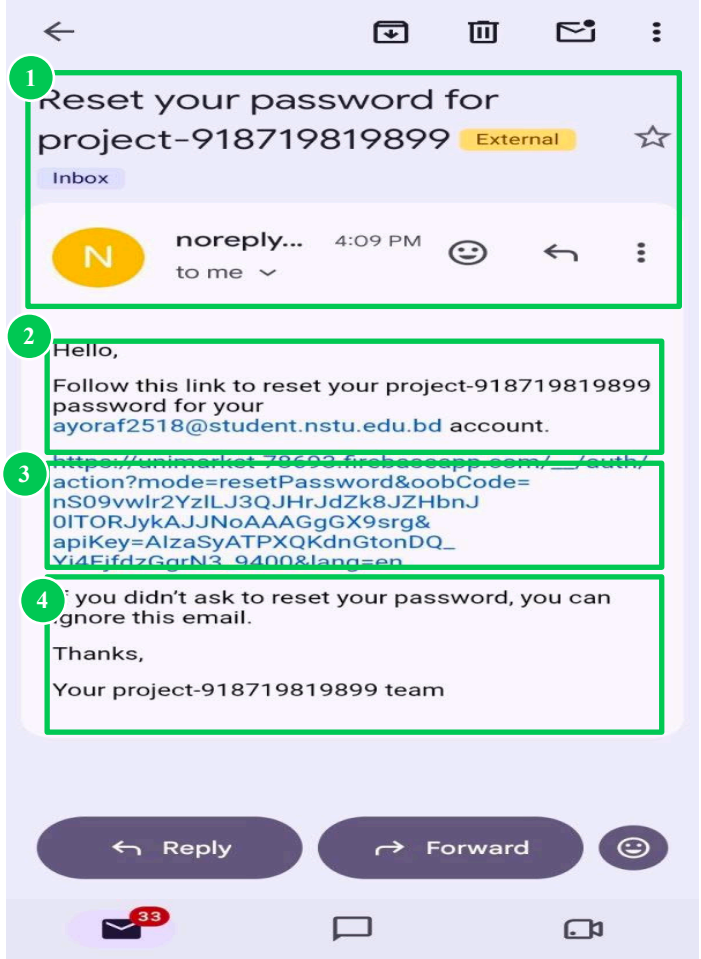


Figure 6.4: Password Reset Confirmation Email

1. Sender Header & Subject: Official automated email header dispatched by the UniMarket Firebase backend (noreply@unimarket...).

2. Target User Account: Displays the verified student email account address associated with the password reset request.

3. Password Reset Action Link: Secure, encrypted HTTPS URL that redirects the student to the dedicated password modification portal.

4. Security Disclaimer Notice: Protective notice instructing users to ignore the communication if the reset request was not self-initiated.

6.1.4 Home Feed, Category Filtering & Search Navigation

The Home Dashboard serves as the central hub of UniMarket, presenting a dynamic catalog of campus goods, category filters, a global search engine, and persistent bottom navigation.

Operational Procedure:

1. Use the Search Bar at the top of the feed to search for specific items (e.g., "Calculus

Book", "Scientific Calculator").

2. Tap any Category Chip (All, Textbooks, Electronics, Dorm Living, Fashion) to filter listings instantly.

3. Scroll through the "Recent Uploads" grid to browse verified products uploaded by fellow students.

4. Utilize the Bottom Navigation Bar to switch between Home, Chat, Post Item (+), Wishlist, and Profile modules.

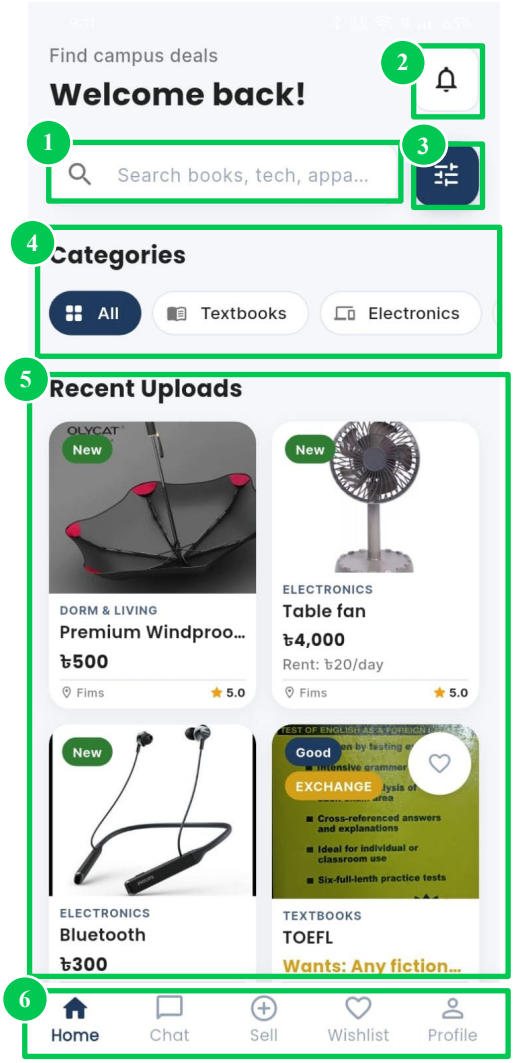


Figure 6.5: Home Dashboard and Category Navigation Feed

1. Search Bar: Real-time text query bar allowing students to search for products by title, category, or relevant keywords.

2. Notification & Cart Shortcuts: Header action icons providing one-tap access to system notifications and the active shopping cart.

3. Category Filter Chips: Horizontal scrolling filter chips enabling instant categorisation across academic books, electronics, dorm essentials, and apparel.

4. Category: It Include all categories of product and help to find product easily.

5. Recent Uploads Feed: Responsive product grid displaying approved items with high-resolution thumbnails, condition badges, prices, and seller locations.

6. Bottom Navigation Bar: Fixed global navigation bar with persistent access to Home, Real-Time Chat, Sell (+), Wishlist, and User Profile.

- **Home** – Shows the main marketplace page where users can browse products and recent uploads.
- **Chat** – Allows buyers and sellers to communicate about products.
- **Sell** – Allows users to upload and sell their products.
- **Wishlist** – Saves favorite products for later.
- **Profile** – Shows the user's personal information, account settings, and activity.

6.1.5 Product Upload (Sell / Rent / Exchange)

Students can list items on the marketplace under three distinct operational modes: direct Selling, short-term Renting, or non-monetary Barter/Exchange. The listing wizard guides the user through multi-step specifications to ensure high-quality catalog data.

Operational Procedure (Sell)

1. Tap the central "+" (Sell) button on the bottom navigation bar.
2. Tap the photo upload area to select clear photos from your mobile device gallery.
3. Enter an accurate, descriptive Product Title.
4. Select the appropriate Category (e.g., Textbooks, Electronics)
5. Select Item Condition (e.g., New, Like New, Good).
6. Choose the Listing Type Sell Only
7. Set the Sell Price
8. Set the stock quantity
9. Provide a detailed description of features, defects, or accessories
10. Tap "Post Item" to submit for admin moderation.

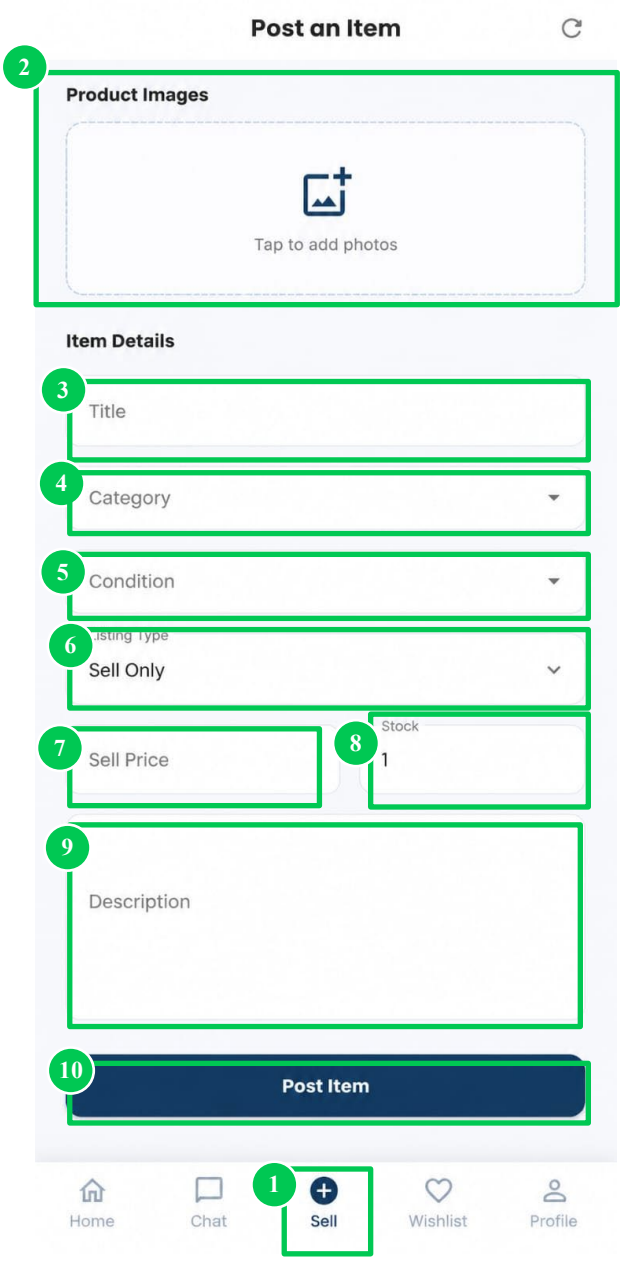


Figure 6.6: Post an Item for sell

Operational Procedure (Rent)

1. Tap the central "+" (Sell) button on the bottom navigation bar.
2. Tap the photo upload area to select clear photos from your mobile device gallery.
3. Enter an accurate, descriptive Product Title.
4. Select the appropriate Category (e.g., Textbooks, Electronics)
5. Select Item Condition (e.g., New, Like New, Good).

6. Choose the Listing Type Rent Only

7. Set the Daily Rent Price

8. Provide a detailed description of features, defects, or accessories

9.Tap "Post Item" to submit for admin moderation.



Figure 6.7: Post an Item for Rent

Operational Procedure (Exchange)

1. Tap the central "+" (Sell) button on the bottom navigation bar.

2. Tap the photo upload area to select clear photos from your mobile device gallery.

3. Enter an accurate, descriptive Product Title.

4. Select the appropriate Category (e.g., Textbooks, Electronics)

5.Select Item Condition (e.g., New, Like New, Good).

6. Choose the Listing Type Exchange Only

7. Give wanted item

8.Give Exchange notes if need

9. Provide a detailed description of features, defects, or accessories

10.Tap "Post Item" to submit for admin moderation.



Figure 6.8: Post an Item for Exchange

6.1.5 Product Details & Item Specification View

Tapping on any product card opens the comprehensive Product Details Screen, which provides high-resolution product imagery, technical specifications, seller reputation, and transaction actions.

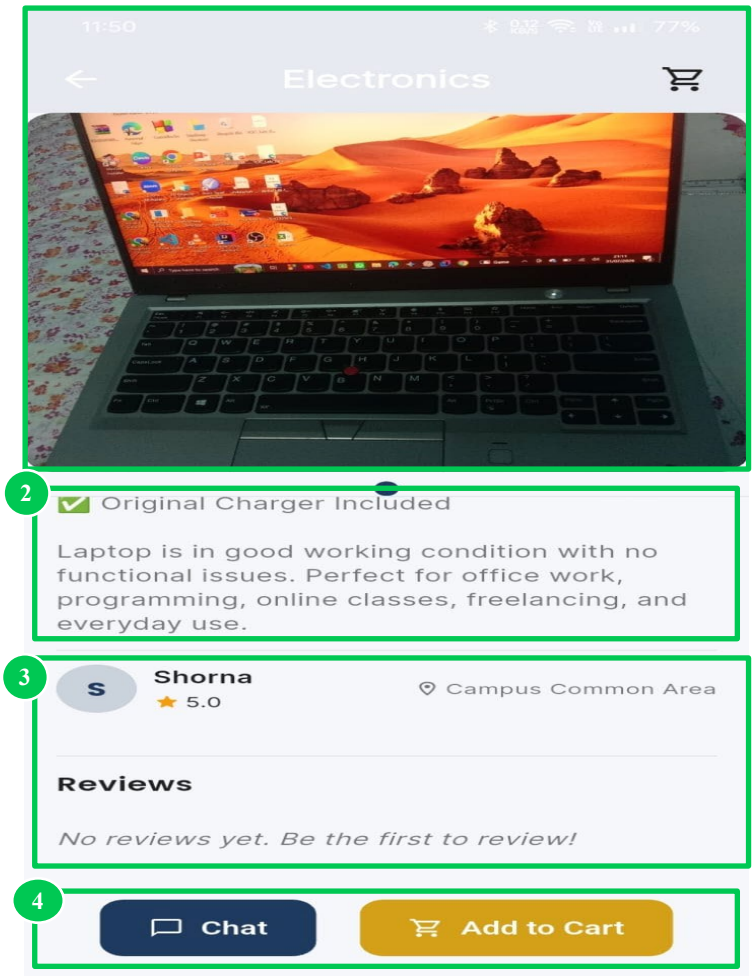


Figure 6.9: Product Details and Specifications Screen

1. Product Image Carousel: Multi-photo interactive carousel displaying detailed visual evidence of the item from multiple viewpoints.

2. Item Description & Specifications: Comprehensive description outlining product history, specifications, warranty status, and reasons for listing.

3. Seller Profile & Rating Card: Verification card showing the seller's identity, academic department, campus meetup zone, and peer review score.

4. Direct Action Buttons: Interactive action controls to start a direct message thread, add the item to the shopping cart, or initiate a rental booking.

6.1.7 Shopping Cart, Checkout & Digital Payment (Buying Flow)

UniMarket features an integrated e-commerce cart, checkout, and digital payment pipeline supporting Cash on Campus Delivery as well as instant digital transactions via the SSLCommerz payment gateway.

Operational Procedure:

1. Tap "Add to Cart" on any product details screen.
2. Open your Cart from the top header to review selected items, adjust quantities, or remove items.
3. Tap "Proceed to Checkout" to review the order breakdown and buyer contact details.
4. Select your preferred payment channel (Digital Payment or Cash on Delivery) and tap "Confirm Order".
5. If digital payment is chosen, complete the transaction via the SSLCommerz portal using bKash, Nagad, Rocket, or Debit/Credit Cards.

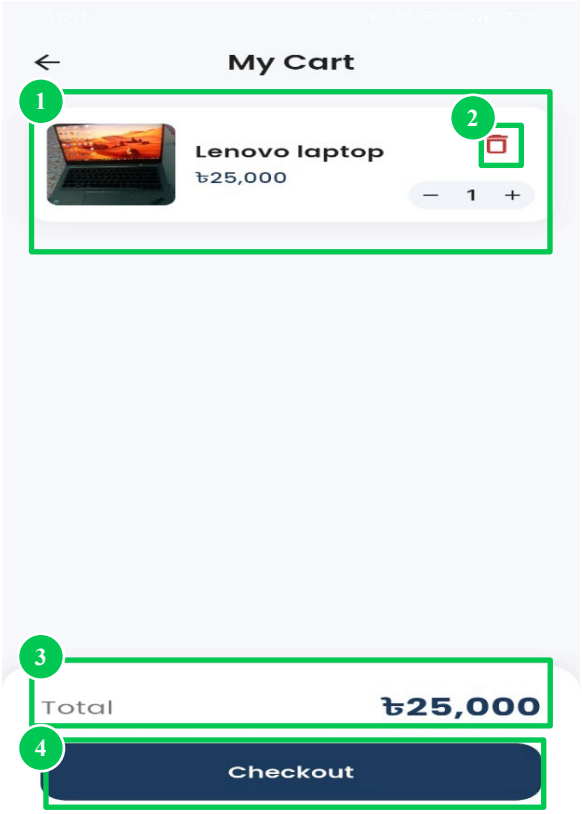


Figure 6.10: Shopping Cart Review Screen

1. Cart Item Card: Item card displaying thumbnail image, product title, unit price, and quantity increment/decrement controls.

2. Item Removal (Delete) Icon: Trash bin control to remove an unwanted item from the active cart session.

3. Grand Total Amount: Real-time calculated subtotal of all selected items in the cart.

4. Proceed to Checkout Button: Action button initiating the delivery details verification and payment sequence.

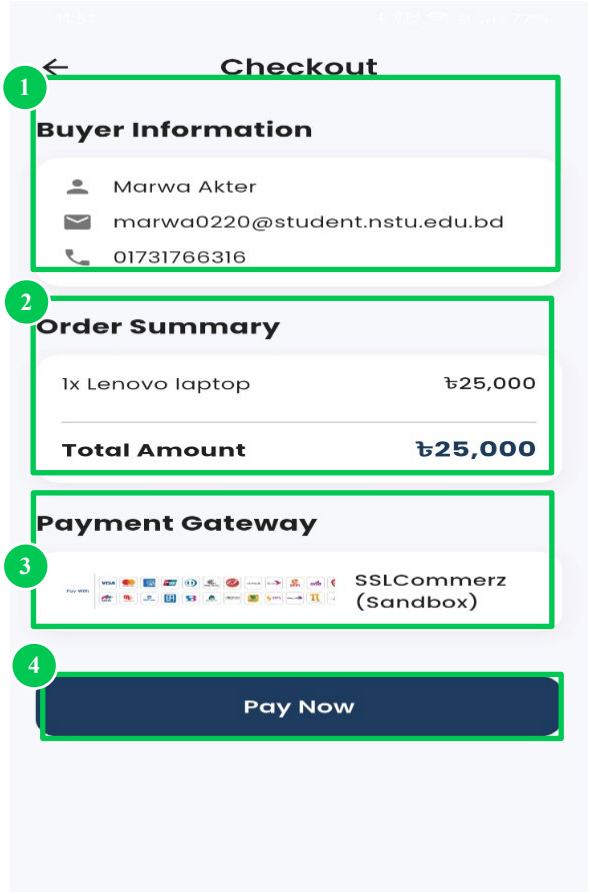


Figure 6.11: Checkout and Order Summary Screen

1. Buyer Information Card: Auto-filled card displaying the student buyer's verified name, institutional email, and contact phone number.

2. Order Summary Breakdown: Line-item invoice outlining all purchased goods, quantities, and the grand total payable amount.

3. Payment Gateway Option: Payment selector allowing the student to choose between digital payment integration or cash handover.

4. Confirm Order / Proceed Button: Final transaction execution button that records the order in Firestore and opens the payment gateway.

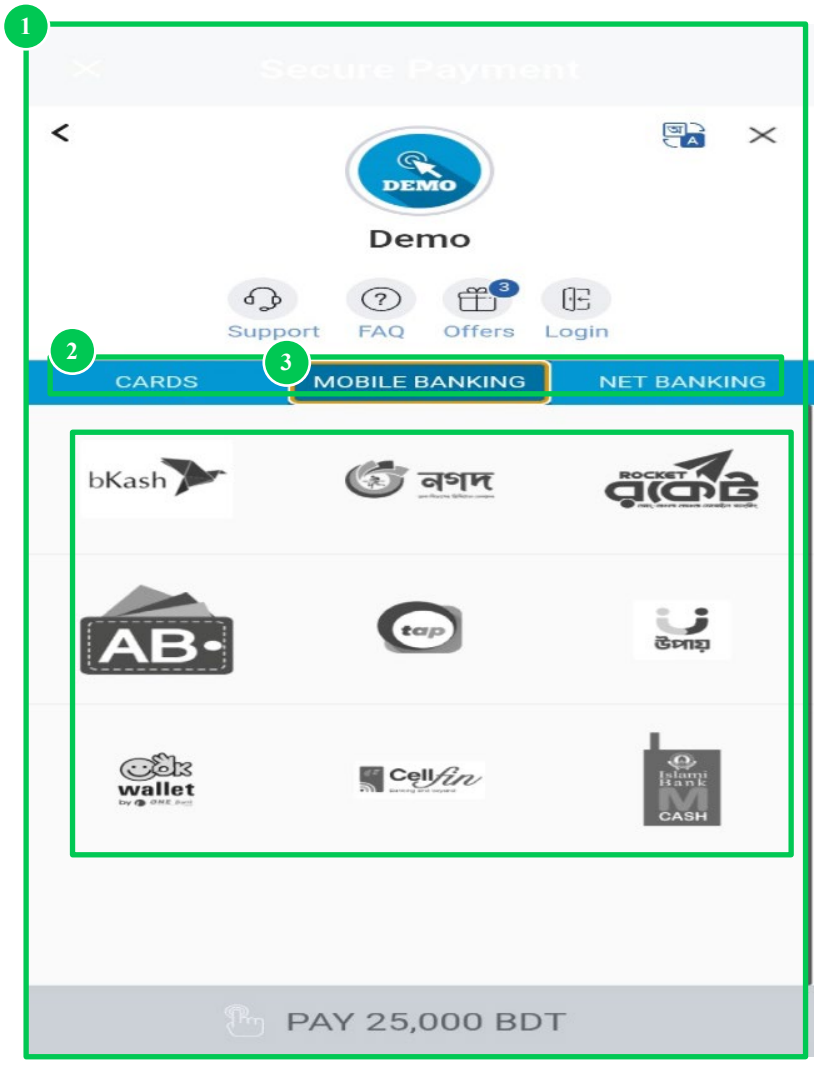


Figure 6.12: Digital Payment Gateway Integration (SSLCommerz)

1. Gateway Header & Payable Amount: Official SSLCommerz gateway interface displaying merchant verification and total payable sum.

2. Payment Channel Tabs: Navigation tabs supporting payment via Cards (Visa, Mastercard), Mobile Banking, and Internet Banking.

3. Mobile Banking Options (MFS): One-tap icons for leading national MFS providers (bKash, Nagad, Rocket, Upay).

6.1.8 Renting System (Booking, Calendar Picker & Cost Calculation)

The Renting module allows students to rent expensive equipment, course books, electronics, or dorm appliances for custom durations without purchasing them outright.

Operational Procedure:

1. Open a rentable product listing and examine the daily rental rate and owner terms.
2. Tap "Rent Now" to launch the interactive calendar booking modal.
3. Tap to select the desired Start Date and End Date for the rental period.
4. Review the system-calculated total rental duration and dynamic aggregate cost.
5. Tap "Confirm Request" to send the binding rental request directly to the item owner.

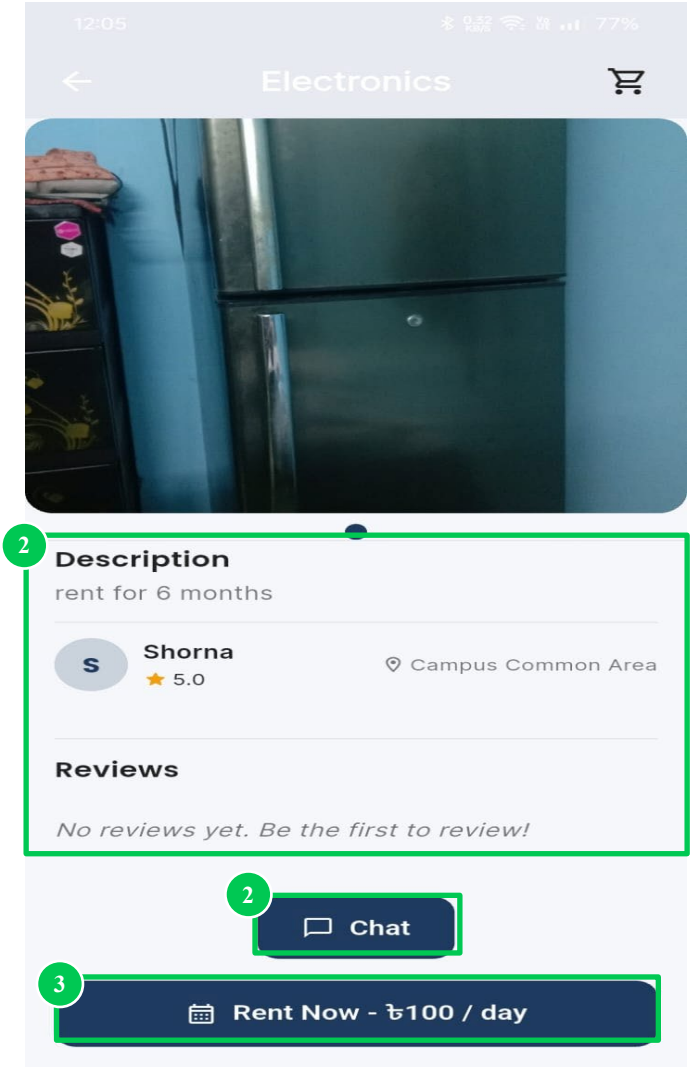


Figure 6.13: Rentable Product Overview Screen

1.Rental Terms & Description: Owner-specified terms outlining maximum allowable rental duration, security deposit, and usage rules.

2.Chat: It is used to chat with product owner

3.Rent Now Action Button: Primary trigger button that initiates the booking workflow and opens the calendar selector.

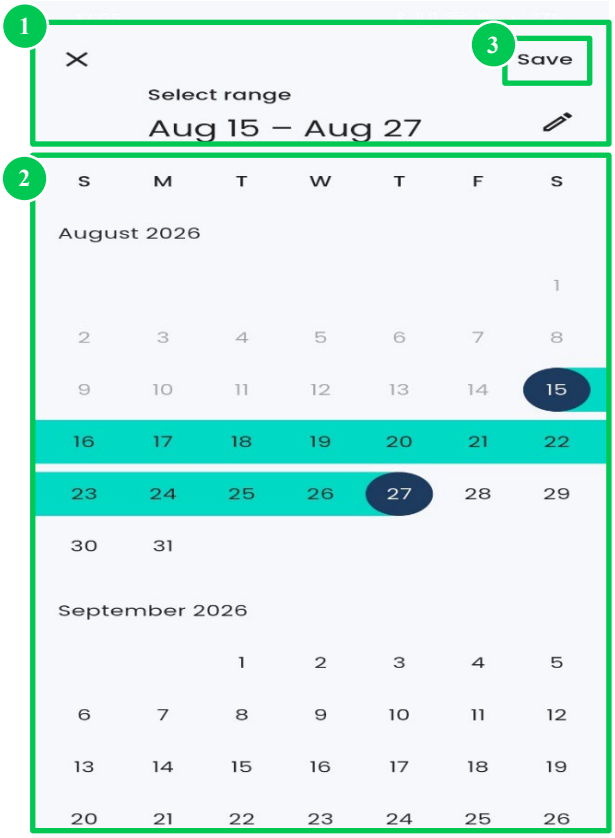


Figure 6.14: Rental Date Range Selector (Calendar Modal)

1. Selected Date Range Display: Header display confirming the currently selected Start Date and End Date range.

2. Interactive Calendar Grid: Visual calendar interface allowing the student to tap and highlight the booking duration.

3. Save / Apply Dates Button: Confirmation button that applies the selected dates and opens the rental calculation modal.

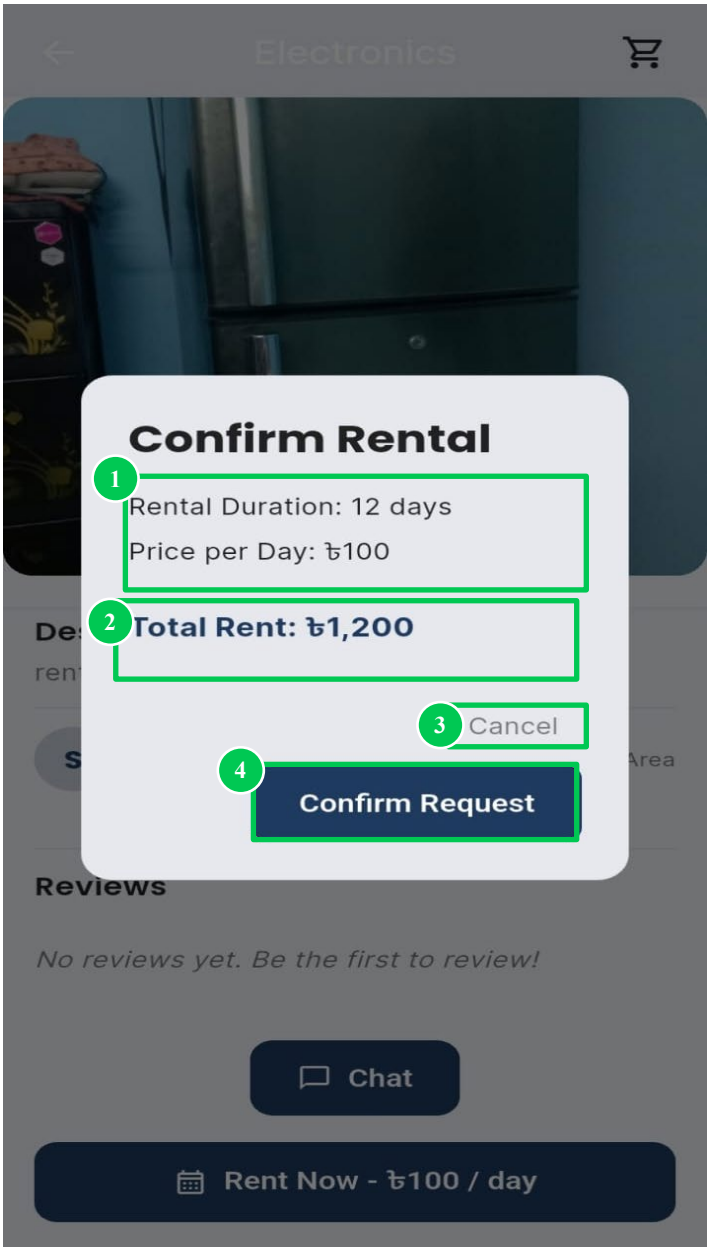


Figure 6.15: Rental Cost Calculation and Confirmation Modal

1. Rental Duration Breakdown: System calculation showing the exact number of days selected for rental (e.g., 12 days).

2. Daily Rate & Total Rent Display: Transparent breakdown multiplying daily rent rate by duration to determine total payable cost (e.g., ৳1,200).

3. Cancel Request Button: Dismissal button to cancel the rental request without committing.

4. Confirm Request Button: Execution button that records the rental request in Firestore and notifies the owner.

6.1.9 Item Exchange System

To encourage a cashless and sustainable campus culture, UniMarket offers a dedicated Exchange module enabling students to trade textbooks, electronics, or equipment directly.

Operational Procedure:

1. Navigate to a product marked for exchange and tap "Send Exchange Request".
2. In the exchange proposal modal, select an item from your own uploaded inventory to offer in trade.
3. Write a brief proposal describing the condition and equivalence of your offered item.
4. Tap "Send Proposal" to submit the swap offer to the listing owner.

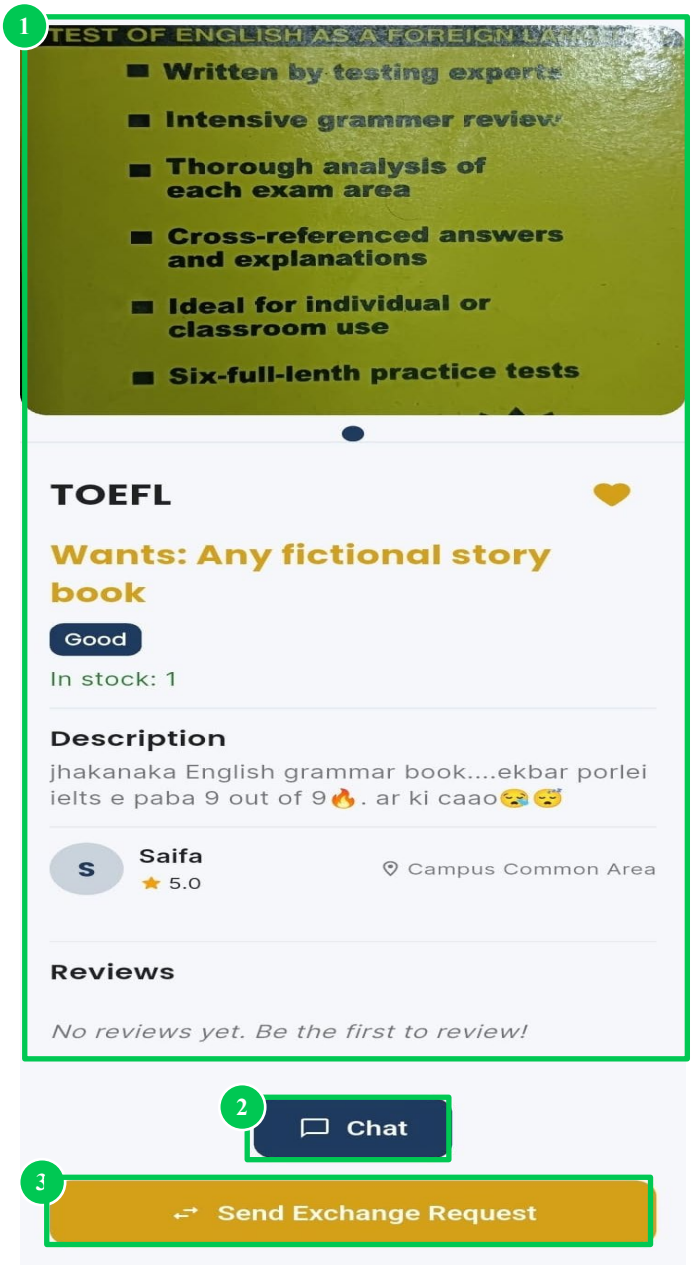


Figure 6.16: Exchangeable Product Overview Screen

1. Product Specifications & Overview: Detailed overview of the product offered for exchange, including model and subject matter.

2.Chat: It is used to chat with product owner2

3. Send Exchange Request Button: Action button that launches the barter proposal modal.

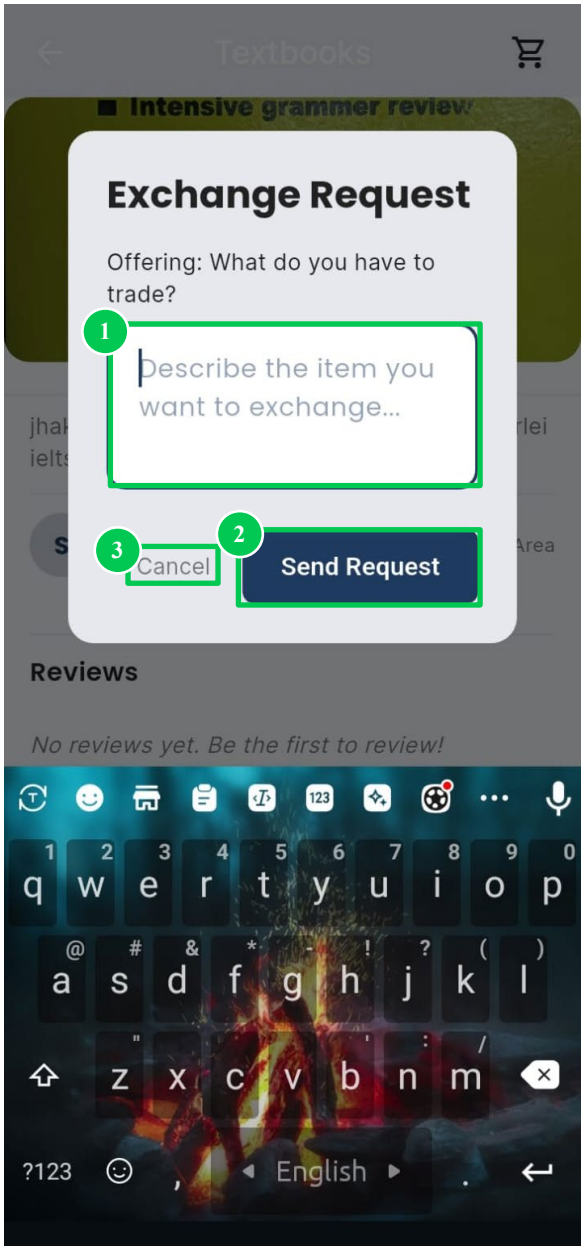


Figure 6.17: Exchange Proposal and Trade Item Selection Modal

1. Proposal Header & Trade Prompt: Informational prompt asking the student what item they wish to offer in return.

2. Send Request Button: Action button that transmits the exchange proposal directly to the owner's inbox and alert center.

3. Cancel: Used for cancel Exchange Request

6.1.11 Real-Time Chat & Direct Messaging System

UniMarket features an integrated real-time chat system powered by Firebase Cloud Firestore, enabling buyers and sellers to negotiate prices, discuss product conditions, and arrange safe campus handovers.

Operational Procedure:

1. Tap the "Chat" icon on the bottom navigation bar to view your active inbox.
2. Tap any conversation thread to enter the live chat room.
3. Type your message and send to communicate instantly with the other party.

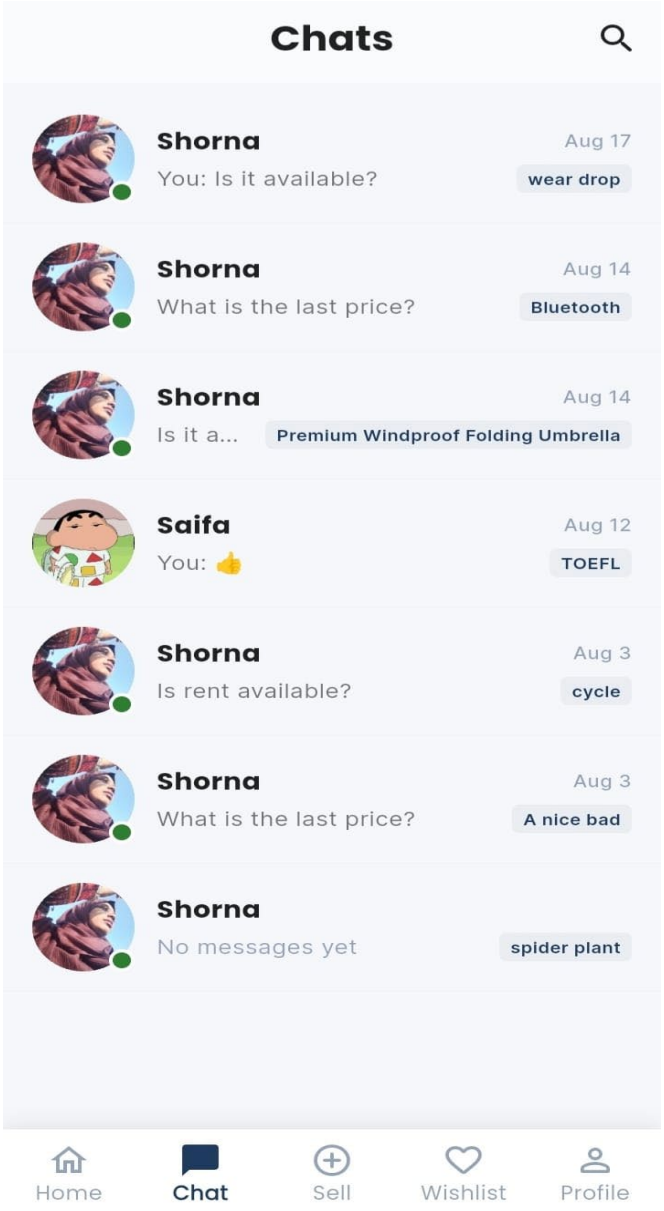


Figure 6.18: Real-Time Chat Inbox and Conversation Threads

6.1.13 User Profile & Account Settings

The User Profile tab allows students to manage personal contact details, review transaction histories across rentals, orders, and exchanges, and safely log out of the application.

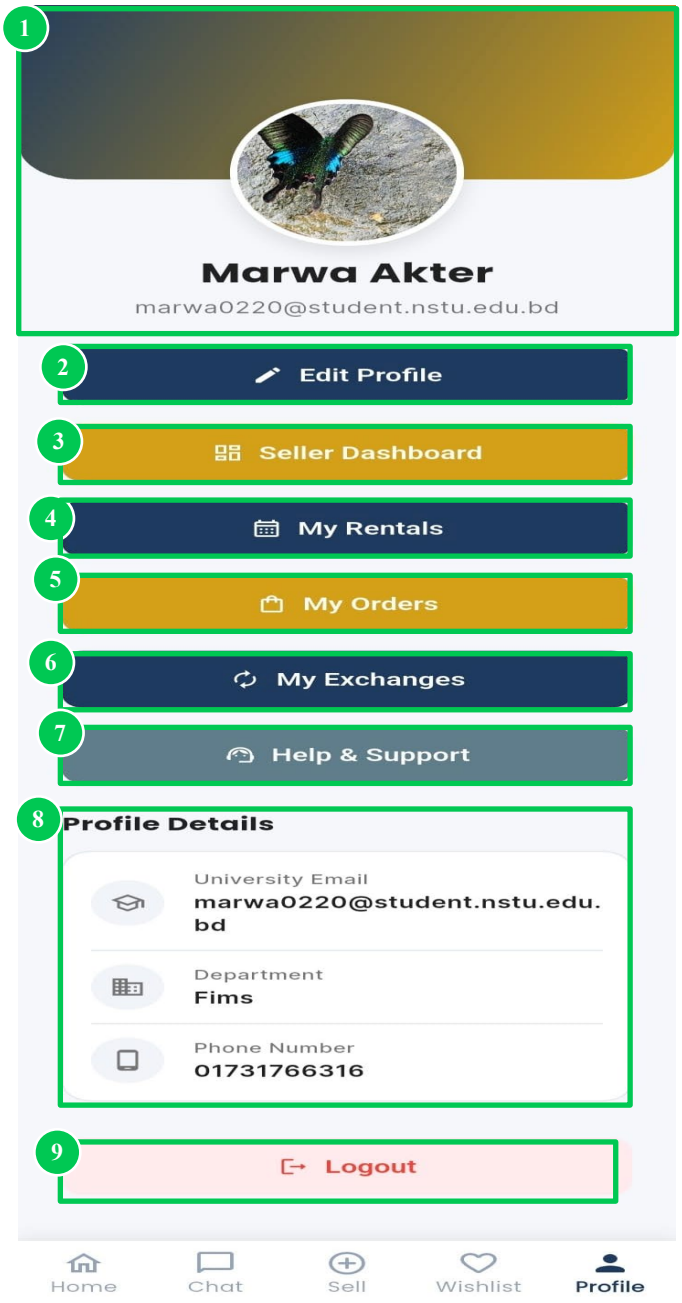


Figure 6.19: User Profile and Account Management Screen

1. Profile Overview

This section displays your basic profile information, including your cover photo, profile picture, full name, and registered university email address.

2. Edit Profile

Click this button to update your personal information, such as your profile picture, phone number, and other account details.

3. Seller Dashboard

Access your dedicated seller portal to manage your listed items, monitor your sales, track earnings, and view your seller activity.

4. My Rentals

Use this option to view and manage the items you have rented through the platform, including your rental history and current rental status.

5. My Orders

Use this option to view and track the status of items you have purchased through the platform.

6. My Exchanges

Click here to view, monitor, and manage your ongoing or previous item exchange requests with other users.

7. Help & Support

Access this section to find answers to frequently asked questions, view helpful guides, or get assistance from customer support.

8. Profile Details

This section displays your account and contact information, including your university email address, department, and phone number.

9. Logout

Click this button to securely sign out of your account and end your current session on the platform.

6.1.10 Seller Dashboard & Inventory Management

Students actively listing goods can monitor their sales metrics, update pricing or descriptions, and manage live inventory via the comprehensive Seller Dashboard.

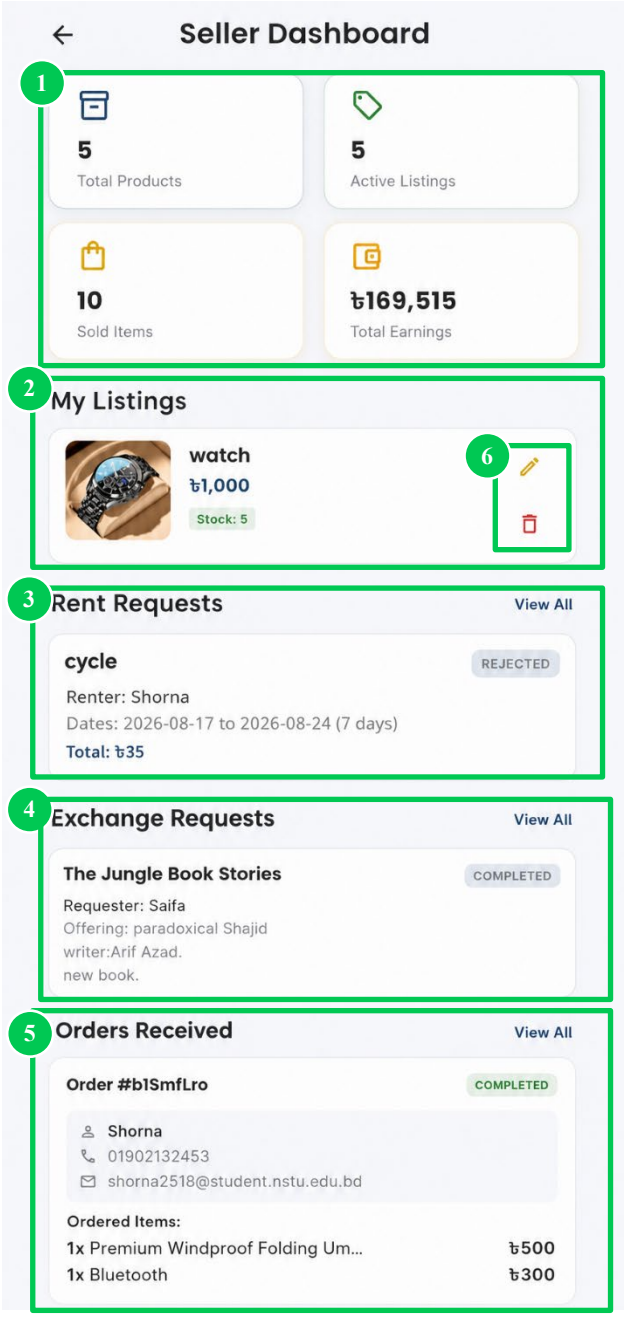


Figure 6.20: Seller Dashboard and Inventory Overview

1. Overview / Summary Cards:

This section provides a quick overview of your selling performance:

- **Total Products:** Displays the total number of products you have added to your shop.
- **Active Listings:** Shows the number of products currently available for buyers to purchase.
- **Sold Items:** Indicates the total number of products you have successfully sold.
- **Total Earnings:** Displays the total revenue generated from completed sales.

2. My Listings:

This section displays the products you are currently selling. It provides key information such as the product image, product name, price, and available stock. The icons marked as 5 allow you to edit or delete a product listing.

3. Rent Requests:

This section displays incoming rental requests from users interested in renting your listed products. It includes the item name, renter's name, requested rental dates, total rental fee, and request status (e.g., REJECTED).

4. Exchange Requests:

This section manages exchange proposals submitted by users who want to trade items with you. It displays the item you are offering, requester's name, item offered in exchange, and exchange status (e.g., COMPLETED).

5. Orders Received:

This section allows you to manage and track purchase orders received from buyers. It includes the Order ID, buyer's contact information (name, phone number, and email), ordered items with their prices, and the current order status (e.g., COMPLETED).

6. Edit & Delete Action Icons: Quick action controls allowing sellers to edit listing details or permanently delete an item

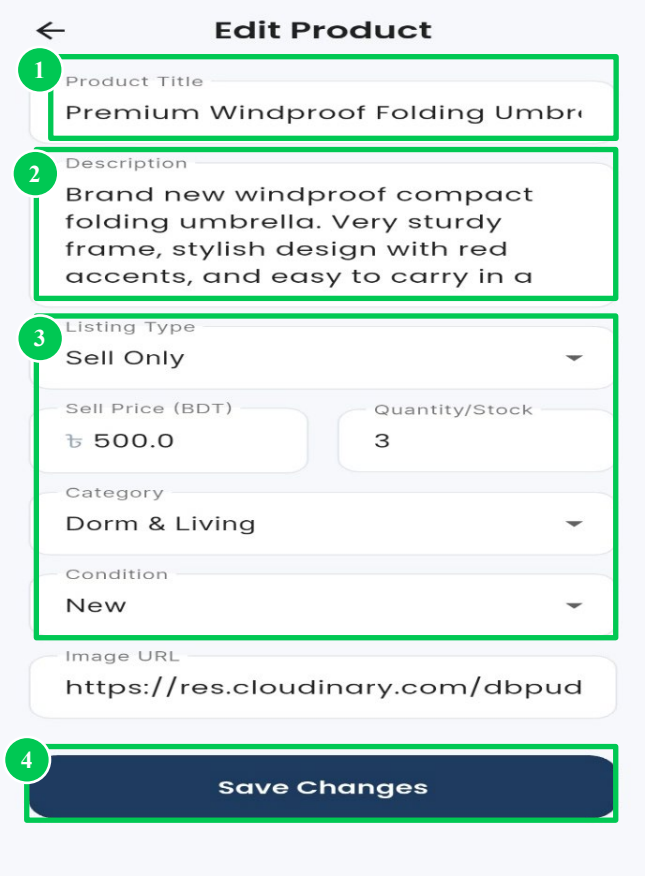


Figure 6.21: Edit Product Listing Details Screen

1. Product Title:

This field allows you to enter or edit the main name of the product you are listing, such as "Premium Windproof Folding Umbrella." The product title is one of the first details buyers see.

2. Description:

This text area allows you to provide detailed information about the product. You can describe its features, quality, dimensions, condition, and other relevant details to help buyers understand the item.

3. Listing Details:

This section contains the key information that determines how your product is listed and sold:

- Listing Type: Specifies whether the item is available for sale, rent, or exchange, depending on the platform's available options.
- Sell Price (BDT): The selling price of the product in Bangladeshi Taka (BDT).

- Quantity/Stock: Indicates the number of units currently available.
- Category: Specifies the category to which the product belongs, such as “Dorm & Living,” helping buyers find it easily.
- Condition: Indicates the current condition of the product, such as New or Used.

4. Save Changes Button:

After updating the required information, click this button to save and apply the changes to your product listing.

6.1.12 Help & Support Ticketing System

To maintain platform trust and resolve transaction disputes or technical issues, UniMarket includes a dedicated Help & Support ticketing module connecting students directly with platform administrators.

Operational Procedure:

1. Go to the Profile tab and tap "Help & Support".
2. View your past ticket history and status badges (Pending, Resolved).
3. Tap the floating "+" (New Ticket) button to open the ticket submission form.
4. Select an Issue Category (e.g., Payment Issue, Bug Report, Item Dispute).
5. Enter an optional Order ID/Product ID and write a detailed description of the problem.
6. Tap "Submit Ticket" to dispatch the issue directly to the admin queue.

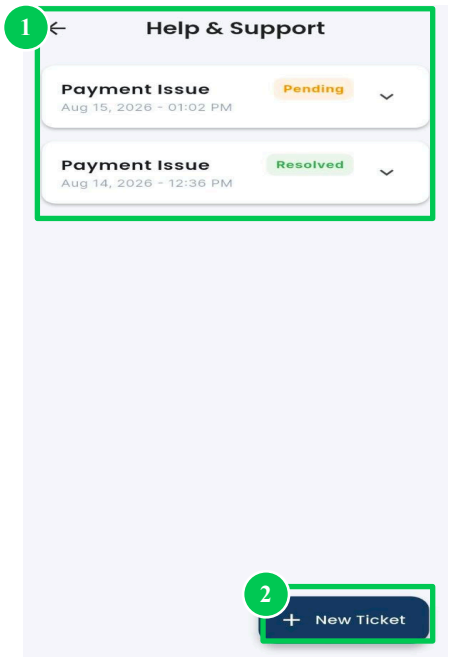


Figure 6.22: Help & Support - Support Ticket History Screen

1. Screen Navigation Header: Top header displaying "Help & Support" title with back navigation to user profile.

2. Support Ticket List Card: Ticket summary card showing issue category, submission timestamp, and real-time status badge (Pending / Resolved).

3. New Ticket Floating Button (+): Floating action button allowing students to initiate a new support ticket complaint.

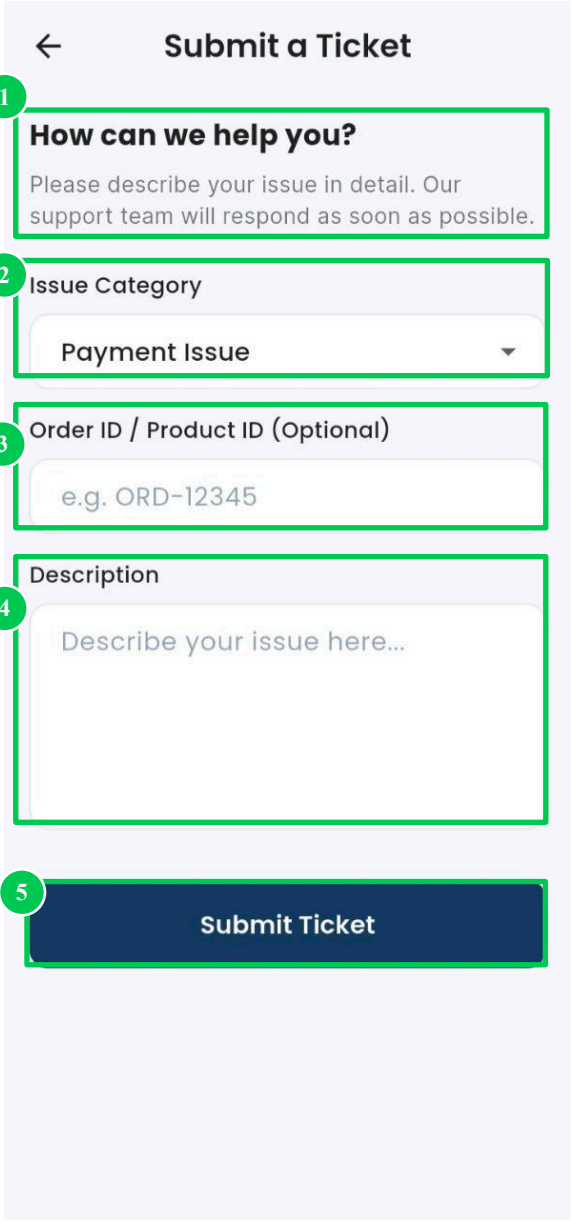


Figure 6.23: Help & Support - Submit a Support Ticket Form

1. Support Guidance Banner: Instructional banner outlining support response expectations and guidelines for reporting issues.

2. Issue Category Dropdown: Selection menu to categorize the ticket under Payment Issue, Product Quality, Bug, or Dispute.

3. Order ID / Product ID Input: Optional reference field linking the ticket directly to a specific transaction or listing.

4. Issue Description Area: Multi-line text area for entering detailed factual explanations of the issue or dispute.

5. Submit Ticket Button: Action button that records the ticket in Cloud Firestore and alerts the platform administrator.

6.2 Admin Application Manual (For Administrators)

The UniMarket Admin Application is a dedicated management console reserved exclusively for platform administrators. It provides centralized controls to monitor marketplace traffic, verify and moderate uploaded products to prevent spam, and resolve student support tickets.

6.2.1 Admin Navigation Drawer

The Navigation Drawer provides instant, centralized access to all administrative modules, allowing platform overseers to transition seamlessly between user accounts, inventory approvals, rental orders, financial payouts, and support tickets.

Operational Procedure:

1. Tap the menu drawer icon located at the top-left corner of the admin app.
2. Review the administrator identity displayed in the header card.
3. Tap any navigation item to switch directly to the desired administrative management screen.
4. Use the "Logout" button at the bottom of the drawer to safely terminate the admin session.

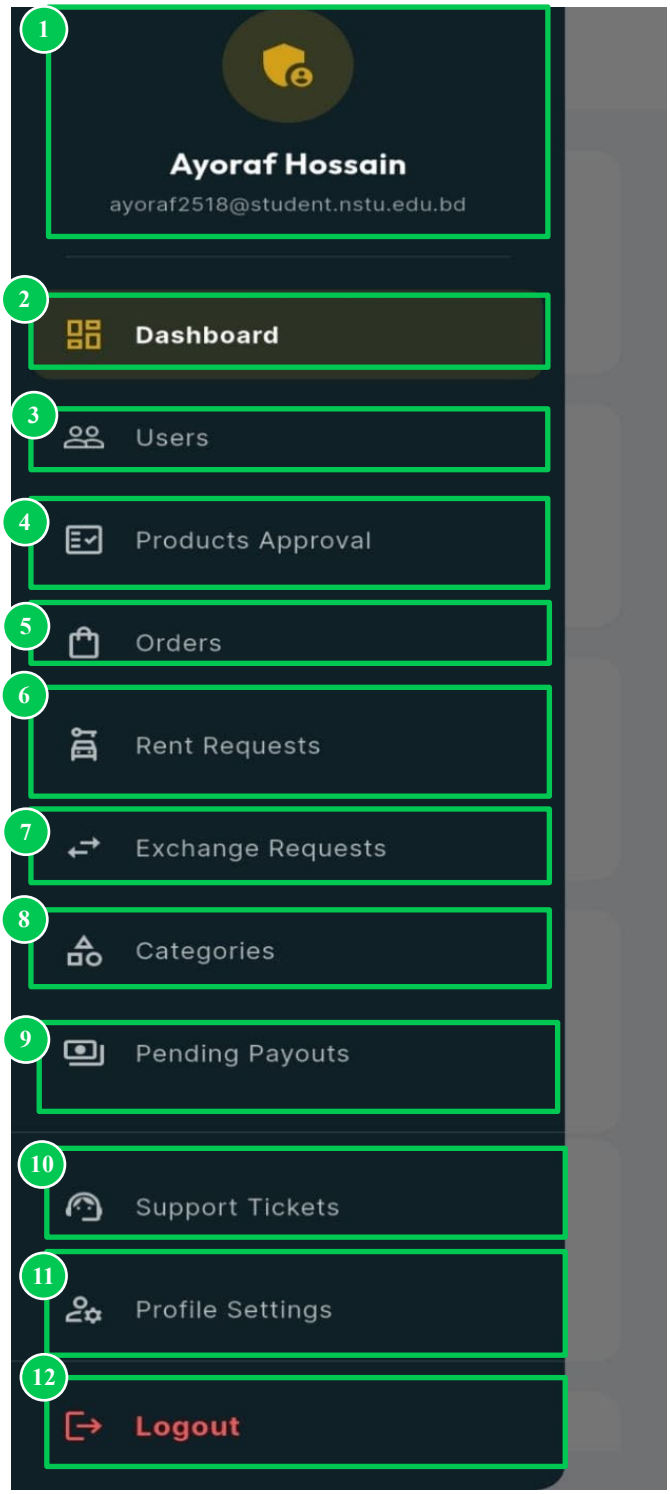


Figure 6.24: Admin Application Navigation Drawer

1. Administrator Profile Header: Displays the authenticated administrator's full name and institutional university email address.

2. Dashboard Overview Menu: Navigates directly to the high-level analytical dashboard displaying vital platform metrics.

3. Users Management: Provides access to the registered student user database for account verification, moderation, and activity monitoring.

4. Products Approval: Direct link to the pending product moderation queue where newly uploaded items are verified before publishing.

5. Orders Management: Management portal to oversee all completed and active purchase transactions across the campus.

6. Rent Requests: Module to monitor, track, and manage ongoing rental bookings and return deadlines.

7. Exchange Requests: Dedicated module to monitor student barter proposals and ensure fair trade compliance.

8. Categories Management: Administrative panel to create, configure, and manage marketplace categories and taxonomies.

9. Pending Payouts: Financial tracking module to verify and process seller payouts and transaction balances.

10. Support Tickets: Access to student grievance tickets and dispute resolution workflows.

11. Profile Settings: Configuration panel to update administrative account settings, credentials, and notification preferences.

12. Logout Control: Secure action button to terminate the administrative session and exit to the login screen.

6.2.2 Admin Statistical Dashboard

The Statistical Dashboard provides real-time visibility into overall marketplace health, displaying high-level operational metrics across student registrations, product inventory, order volumes, platform revenue, and rental/exchange requests.

Operational Procedure:

1. Open the "Dashboard Overview" from the navigation drawer.
2. Inspect top-level cards for Total Users and Total Products currently live in the catalog.
3. Track transactional performance across Total Orders and Total Platform Revenue.
4. Monitor ongoing campus engagement via Rent Requests and Exchange Requests counters.

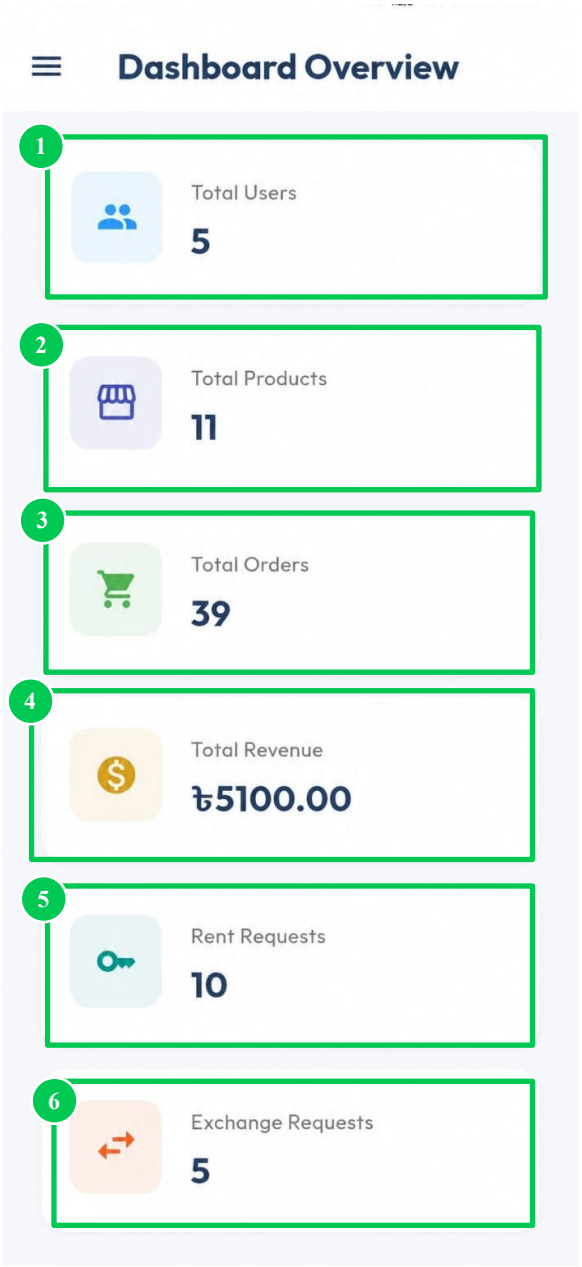


Figure 6.25: Admin Statistical Dashboard Overview

1. Total Users Metric Card: Real-time counter displaying total number of registered student accounts across the university campus.

2. Total Products Metric Card: Real-time counter showing the total number of approved and active inventory listings.

3. Total Orders Metric Card: Real-time summary of all completed and active purchase transactions.

4. Total Revenue Metric Card: Aggregate gross transaction volume processed across the platform in BDT.

5. Rent Requests Metric Card: Real-time count of active and completed rental bookings across the campus.

6. Exchange Requests Metric Card: Real-time counter of submitted and active barter/item trade proposals.

6.2.3 Product Moderation & Quality Verification (Approve / Reject)

To maintain high catalog standards and prevent prohibited items, every newly listed product enters a pending moderation queue where administrators review listing quality before making it visible publicly.

Operational Procedure:

1. Open the "Products Approval" module from the admin navigation drawer.
2. Review product details: listing type, category, condition grading, price, uploaded images, and seller identity.
3. Tap "Approve" to publish the item immediately to the global student Home Feed.
4. Tap "Reject" if the product violates campus regulations, notifying the seller with an explanation.

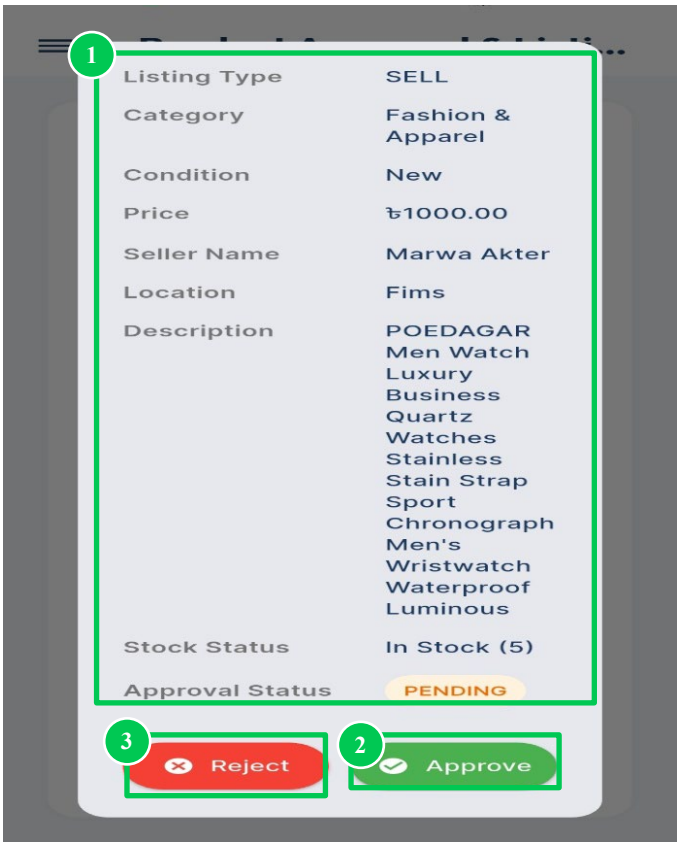


Figure 6.26: Product Moderation and Verification Screen

1. Product Details Panel: Panel presenting listing type (Sell/Rent/Exchange), category, condition, price, and uploaded images for verification. Displays the student seller's name and the exact date/time of listing submission.

2. Approve Product Button: Action button that sets listing status to "Approved", making the item publicly browsable in the marketplace feed.

3. Reject Product Button: Action button that marks the item as "Rejected" and sends moderation feedback to the seller.

6.2.4 Support Ticket Resolution & User Dispute Management

Administrators manage incoming user complaints, payment discrepancies, and transaction disputes via the centralized Support Tickets module.

Operational Procedure:

1. Navigate to the "Support Tickets" section from the navigation drawer.
2. Review the ticket queue displaying pending issues, user names, and timestamps.
3. Tap an open ticket to inspect user details, associated Order IDs, and the full description of the dispute.
4. Type an official resolution response in the reply text field and tap "Mark as Resolved" to close the ticket.

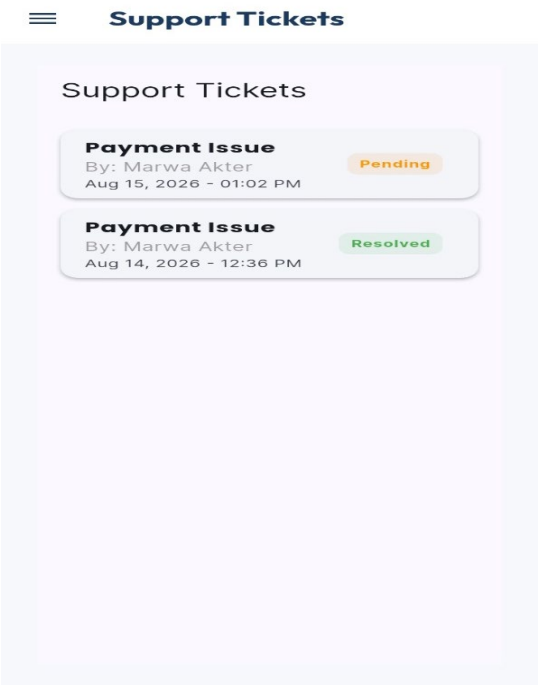


Figure 6.27: Admin Support Tickets Queue Screen

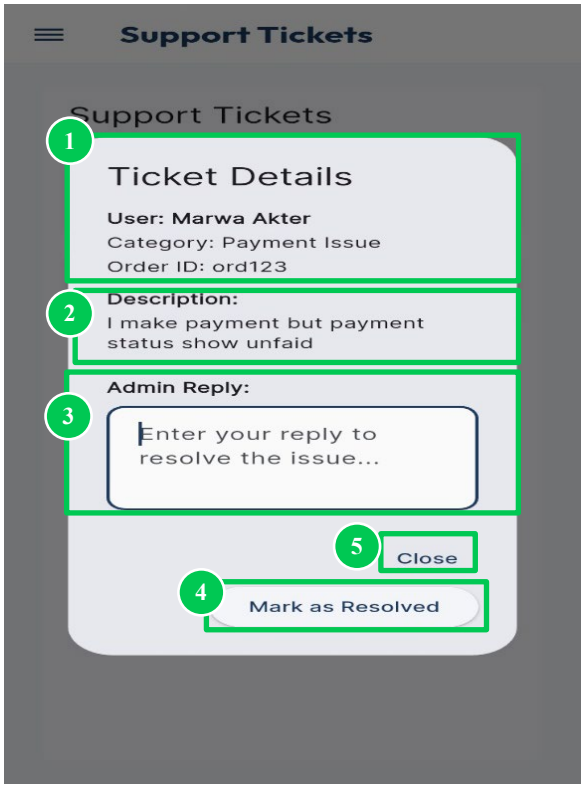


Figure 6.28: Admin Support Ticket Resolution and Reply Screen

1. Ticket Metadata Card: Card showing submitting student name, issue category, timestamp, and linked Order ID.

2. User Complaint Description: Unedited text submitted by the student detailing the problem, transaction issue, or grievance.

3. Admin Resolution Reply Input: Text field for typing the official resolution, instructions, or administrative action taken.

4. Mark as Resolved Button: Action button that transmits the resolution to the student's support interface and marks the ticket as Resolved.

5. Close Button: Clicking this button dismisses the current window and takes you back to the main Support Tickets list **without** saving any draft replies, sending a message, or changing the status of the ticket. It acts as a "Cancel" or "Go Back" action.

7. Input and Output Description

This section outlines the flow of data for the major functions of the UniMarket system, detailing the specific inputs required from the user, the background processing performed by the system, and the final expected output.

7.1 User Registration and Authentication

- **Input Required:** The user inputs their Full Name, University Email Address, Phone Number, and a secure Password. Alternatively, an OAuth token is provided if using Google Sign-In.
- **Processing:** The system securely transmits these credentials to Firebase Authentication over an encrypted network. Firebase validates the email format, checks for duplicates, hashes the password, and generates a unique user ID (UID). A corresponding user document is also created in the Firestore database.
- **Expected Output:** If valid, the system outputs a success message and redirects the user to the Home Dashboard. If invalid, an error message (e.g., "Email already in use") is displayed.

7.2 Product Upload (Sell/Rent/Exchange)

- **Input Required:** Product Images (from gallery), Title, Price, Rent Price (if applicable), Category, Condition, Stock Quantity, Listing Type, and a brief Description.
- **Processing:** The application uploads the selected images to Firebase Cloud Storage, which generates secure download URLs. The system then bundles these URLs with the text inputs into a structured JSON document and saves it to the products collection in Firestore with an initial approvalStatus set to "pending".
- **Expected Output:** The seller sees a "Product Uploaded Successfully" confirmation. The product instantly appears on the Admin Dashboard for review, but remains hidden from the public feed.

7.3 Rent Request System

- **Input Required:** The buyer selects a rental Start Date and End Date from a visual calendar picker.
- **Processing:** The system calculates the total number of rental days by finding the mathematical difference between the two dates. It then multiplies these total days by the product's defined rentPricePerDay. A rent request document is created and pushed to the database.
- **Expected Output:** The calculated "Total Rental Cost" is displayed dynamically on the screen before submission. Upon confirmation, the seller receives a real-time notification containing the request details.

7.4 Order Placement / Checkout

- **Input Required:** The buyer inputs their Delivery Details/Location and selects a Payment Method (Cash on Delivery or Digital Payment).
- **Processing:** The system iterates through all the items currently in the cart to calculate the grand total price. It generates a new orders document in the database and automatically decrements the stockQuantity of the purchased items in the products collection to prevent overselling.
- **Expected Output:** The UI displays an "Order Successful" confirmation page. The buyer's cart is automatically cleared, and the respective sellers instantly receive a "New Order" notification.

7.5 Admin Product Moderation

- **Input Required:** The Administrator taps either the "Approve" or "Reject" button on a pending product card.
- **Processing:** The system updates the approvalStatus field of the specific product document in Firestore from "pending" to "approved" or "rejected". If rejected, it may also execute a delete command to remove the product entirely.
- **Expected Output:** If approved, the product instantly becomes visible on the global Home Feed of all users. In both cases (approval or rejection), a real-time notification is sent to the seller's device informing them of the decision.

7.6 Support Ticketing System

- **Input Required:** The user inputs an Issue Category (e.g., Bug, Dispute), a Subject line, and a detailed Description of their problem.
- **Processing:** The system creates a support_tickets document linking the user's ID to the complaint. When an admin replies, the system updates the adminReply field and changes the status to "Resolved".
- **Expected Output:** The user sees their submitted ticket in their profile. Once resolved by the admin, the user receives an alert and can read the admin's feedback directly within the app.

8. Error Messages and Troubleshooting

This section outlines common errors or problems that users may encounter while using the UniMarket application, along with their possible reasons and solutions.

Table 8.1: Error Messages and Troubleshooting

Problem	Possible Reason	Solution
Unable to log in	Incorrect email/password combination or unregistered email.	Check your credentials carefully and try again, or use the "Continue with Google" option.
Product images not uploading/loading	Slow or unstable internet connection disrupting Firebase Storage.	Ensure your device is connected to a stable Wi-Fi or 4G network and try again.
Uploaded product not showing on Home Feed	The product is still pending Admin approval or was rejected.	Wait for the administrator to approve the listing. Check your notifications for rejection reasons.

"Items no longer available" error when adding to cart	The requested quantity exceeds the seller's current stock limit.	Reduce the item quantity to match the available stock, or contact the seller via Chat.
Notifications not working / Blank Screen	Firebase Background Services may be restricted by the phone's OS.	Go to your phone settings, clear the app's cache, and ensure background data usage is allowed.
App crashes or freezes randomly	Insufficient RAM or outdated application version.	Clear background apps to free up RAM, or reinstall the latest .apk version of UniMarket.

9. Frequently Asked Questions (FAQ)

This section provides answers to common questions that users might have regarding the functionalities and policies of the UniMarket application.

Q1. How can I create an account on UniMarket?

Answer: You can create an account by tapping the "Sign Up" link on the welcome screen and filling in your details, or by simply using the "Continue with Google" option for instant registration. For detailed steps, refer to Section 6.1.1.

Q2. What should I do if I forget my password?

Answer: Use the "Forgot Password?" option available on the login page. A secure password reset link will be sent to your registered university email address via Firebase.

Q3. I uploaded a product to sell, but I cannot see it on the Home Feed. Why?

Answer: To maintain a secure and spam-free ecosystem, UniMarket requires all newly uploaded products to undergo an Admin moderation process. Your product will appear on the global feed as soon as an administrator reviews and approves it.

Q4. How does the Rent feature work?

Answer: If a seller has listed an item for rent, you can click the "Rent Request" button on the product page. A calendar will appear allowing you to select your Start and End dates. The app will automatically calculate the total cost, and you can send the proposal to the seller.

Q5. Can anyone outside our university use this application?

Answer: No. UniMarket is designed as a hyper-local, campus-centric ecosystem. Only verified university students with valid student emails can register and participate in transactions, ensuring a high level of trust and safety.

Q6. What should I do if a buyer/seller scams me or if I face a transaction dispute?

Answer: UniMarket has a dedicated Help & Support module. You can navigate to your Profile, open a "Support Ticket", and describe your issue. Our administrators monitor these tickets in real-time and will intervene to resolve the dispute.

Q7. Is there any extra charge for proposing an item exchange?

Answer: No, the exchange module is completely free. It is designed to promote a sustainable culture of resource sharing among students without involving any monetary transactions.

10. Logout and Exit Procedure

To ensure data security, especially if using a shared device, users should safely log out of their accounts when they are done using the application.

- **How to Log Out:**
    1. Navigate to the **Profile** tab on the bottom navigation bar.
    2. Scroll to the very bottom of the profile screen.
    3. Tap the **"Log Out"** button.
    4. Confirm the prompt (if any). The system will securely terminate your active Firebase session and redirect you to the Welcome/Login screen.
- **How to Exit the Application:**

Once logged out, or if you simply wish to close the app while keeping your session active, you can exit the application by swiping it away from your device's "Recent Apps" menu or by pressing your phone's physical Home/Back button.

11. Limitations and Precautions

While operating the UniMarket system, users must be aware of the following operational limitations and safety precautions:

Limitations:

- **No Physical Delivery Logistics:** The platform acts as a digital matchmaking and order-processing system. UniMarket does not provide dedicated delivery personnel; buyers and sellers must manually coordinate a physical meetup to exchange the product.
- **Condition Verification:** The system cannot technically verify the physical condition of an item (e.g., if a rented calculator is fully functional). Transactions rely on the honesty of the uploaded images and mutual trust.

Precautions:

- **Secure Meetups:** Always choose a public, well-lit area within the university campus (such as the library or cafeteria) when physically meeting someone to hand over a product or cash.
- **Account Security:** Never share your UniMarket account password or Google credentials with anyone in the chat. Administrators will never ask for your password.
- **Inspect Before Payment:** When receiving a rented or bought item, physically inspect the product to ensure it matches the description before completing the payment or handover.

12. Contact / Support Information

If users encounter technical bugs, face transaction disputes, or need administrative assistance, they can reach out to the support team through the following channel:

In-App Support Ticket:

The fastest way to get help is to use the built-in ticketing system. Navigate to **Profile > Help & Support**, tap the '+' icon, and describe your issue. An administrator will reply to your ticket within 24 hours.

Supervisor Approval:

Project Title:

UniMarket: A Student Marketplace for Buying, Selling, Renting, and Exchanging Items

Group Members:

Name	**Student ID**
Ayoraf Hossain	ASH2225001M
Saifa Sikder Shauly	BKH2325013F
Jannatul Ferdous Shorna	JSH2325035F

I have reviewed the User Manual Report of the above-mentioned SPL-2 project.

Supervisor

Tasnim Rahman

Assistant Professor,

IIT, NSTU

_____

Supervisor's Signature

16 August, 2026